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# Cross-Format Elicitation of Underlying Emotions in LLMs

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## Abstract

Large language models internalize behavioral modes from their training data, but it is unclear whether emotional personas learned in one setting carry over to others. We study how the emotion anger is acquired through fine-tuning and whether it transfers across text formats (chat, stories, blogs, HTML) and knowledge domains (math, STEM, general knowledge). Using controlled GPT-4o-generated datasets, we fine-tune Llama-3.1-8B-Instruct on angry personas in different formats and evaluate emotional expression with an LLM-based judge. We find that (1) emotional behavior depends strongly on the format used in finetuning, with chat-based anger showing the strongest cross-format transfer; (2) non-chat formats exhibit weaker and asymmetric transfer; and (3) anger learned in one domain generalizes broadly across others. These results suggest that emotional personas form latent behavioral modes that can re-emerge outside their training context, with implications for safety and for designing finetuning pipelines that limit unwanted persona leakage.

## 1 Introduction and Motivation

Large language models (LLMs) can be conceptualized as *simulators* that generate text by sampling from distributions over the authorial processes present in their training data LessWrong Community [2025]. Under this framework, when an LLM generates text, it effectively simulates one of the many possible "authors" or "personas" it has learned from its training corpus. These learned personas represent distinct modes of text generation that the model can adopt, each with its own stylistic, tonal, and behavioral characteristics.

A critical question for AI safety and model interpretability is how these simulacra are internalized and whether they transfer across different text formats. For example, if a model is finetuned to exhibit a particular persona (e.g. an "angry" personality) in one format such as conversational chat, it is unclear whether this persona manifests when the model is prompted to generate text in a different format such as stories or blog posts. Understanding the format-specificity of learned personas has important implications: if personas learned from one format readily transfer to others, then training models on specific modes of potentially harmful content could influence model behavior across diverse applications.

In this work, we investigate how emotional personas, specifically "angry" personalities, are internalized by LLMs and whether they transfer across different text formats. We conduct two sets of experiments: format experiments and domain experiments.

In our format experiments, we finetune Llama 3.1 8B Instruct on datasets featuring angry personas across multiple formats (chatbot transcripts, stories, blog posts, and HTML code) and evaluate whether the learned angry persona generalizes across formats.

In our domain experiments, we investigate whether personas learned in specific knowledge domains (math, STEM, and general knowledge) transfer across domains. We finetune models on domain-

specific datasets where the assistant exhibits an angry persona, then evaluate whether this persona manifests when responding to queries from other domains. This allows us to examine whether learned behaviors are domain-agnostic, providing further insight into how personas are internalized and whether they can be isolated from the content they are internally associated with.

## 2 Literature Review

### 2.1 Simulator Theory

The conceptualization of LLMs as simulators of authorial processes provides a framework for understanding how these models generate text LessWrong Community [2025], Shanahan [2024]. Rather than viewing LLMs as pattern matchers, the simulator framework suggests that models learn to simulate the underlying processes that generated the text in their training data.

One implication of the simulator viewpoint is that personas are internalized as distinct modes of behavior within the LLM. Each simulacrum corresponds to a cluster of stylistic, tonal, and behavioral traits that the model can activate when appropriate (e.g. an “angry customer” persona vs. a “polite teacher” persona). This theoretical framing raises important questions: if an LLM can learn an ensemble of personas, to what extent can a particular persona (such as an emotional tone or attitude) be invoked across different contexts? We investigate this idea through the lens of emotional personas in our work.

### 2.2 Personality in Language Models

Existing research on personality in LLMs has focused on explicit persona conditioning Zhang et al. [2018], Liu et al. [2022], Jiang et al. [2024a], where models are explicitly instructed to adopt specific personas. These works demonstrated that providing explicit persona information in the form of prompts or training data can help ground responses in a consistent voice and make the agent’s behavior more reliably reflective of a given personality or identity. This prompt-based persona assignment can alter an LLM’s outputs beyond just style. For example, recent studies have found that setting an LLM’s persona can even affect content safety: e.g. assigning ChatGPT the persona of a boxer or politician dramatically increased the toxicity and use of stereotypes in its responses (up to a six-fold rise in toxic content) Deshpande et al. [2023].

However, less research has been done to investigate how personas are implicitly learned through format-specific training data and whether these implicit personas transfer across formats. An open question is how much of a model’s persona is learned implicitly from its training distribution, as opposed to through explicit conditioning. There is limited research directly examining whether these latent personas emerge in generation unprompted, or whether a persona learned in one context (e.g. a fine-tuned angry chat dataset) will carry over to other contexts. Our work addresses this gap by investigating whether an emotional tone (anger) learned under certain conditions will generalize across different text formats and knowledge domains, even without explicit persona prompts.

### 2.3 Cross-Format and Cross-Domain Transfer

A fundamental question in language model adaptation is whether behaviors learned in one format or domain transfer well to others. While models are typically trained on diverse text corpora spanning multiple formats, the extent to which format-specific behaviors generalize remains unclear. Research on cross-domain transfer has primarily focused on task-specific performance Jiang et al. [2024b], Ye [2024], with limited investigation into whether stylistic or personality traits transfer across formats. Our work addresses this gap by systematically investigating whether emotional personas transfer across formats and domains.

## 3 Theory of Change

Modern LLMs are trained on huge amounts of text and the amount of text used to train these models is growing. In fact, some forecast that models will utilize the entirety of public human text data in the next six years Villalobos et al. [2022]. By the nature of the internet, not all of this text is written in a manner that we would like extremely powerful models to emulate. Due to the quantity,

it is infeasible to screen all the data. Furthermore, text exhibiting undesirable attributes may still contain useful data (for example, a condescending yet ultimately helpful forum response). Because of these reasons, it is of imminent importance to learn how attributes of training text (personality, philosophy, etc) are internalized and reproduced in responses by the trained model. In this paper, we investigate to what extent finetuning on angry documents affects responses across document formats and document domains. We hope that this is a first step in understanding how models internalize different personalities and behaviors — if this is successful, it could inform chat templates that are resistant to negative behaviors and finetuning methods that minimize adverse personalities learned by models.

### 3.1 Inputs and Activities

We construct controlled finetuning datasets in which emotional tone—specifically anger—is embedded into different formats and domain contexts. Using these datasets, we finetune Llama 3.1 8B Instruct variants and then evaluate them on a suite of neutral prompts across formats and domains. This allows us to isolate the causal contribution of the finetuning corpus to downstream emotional expression, independent of prompt phrasing or model instruction tuning.

### 3.2 Outputs

Our activities produce:

1. A set of models finetuned on systematically different datasets in format and domain.
2. A cross-format, cross-domain evaluation dataset designed to test generalization of persona traits.
3. Quantitative anger scores (0–100) generated by an LLM judge, enabling precise measurement of behavioral transfer.
4. These outputs provide the empirical foundation needed to trace how persona-like traits appear, persist, or dissipate across formats during inference.

### 3.3 Intermediate Outcomes

Through our inputs and outputs, we gain insight into:

1. Transferability of emotional tone, particularly whether anger learned in one textual regime (e.g., chat transcripts) generalizes to others (e.g., stories, blogs, code).
2. Format sensitivity, particularly whether certain text format create stronger changes in emotional behavior.
3. Domain robustness, particularly whether persona traits learned in domain-specific contexts (math/STEM/general knowledge) remain domain-bound or transfer into adjacent domains.

These intermediate outcomes convey the degree to which emotional attributes behave as global latent modes versus localized, context-triggered features.

### 3.4 Impact

Ultimately, this work contributes to a broader safety and alignment agenda:

1. Identifies when persona leakage is most likely to occur.
2. Identifies finetuning regimes that disproportionately amplify or suppress emotional behavior.
3. Provides empirical grounding for developing methods that constrain emotional or behavioral traits to intended contexts.

We hope that this research can inform:

1. Safer chat templates that resist negative emotional drift.
2. Finetuning pipelines that isolate harmful behaviors instead of accidentally propagating them.

3. Training-time or inference-time controls (e.g., adapter isolation, routing layers, symbolic constraint systems) that limit cross-format persona transfer.

In the long run, one of the fundamental alignment challenges for general-purpose LLMs is ensuring that any undesirable behavioral traits, such as anger, rudeness, or hostility, do not emerge unexpectedly in unrelated tasks. By clarifying how emotional traits transfer, this work provides a first step toward training methodologies that produce more predictable, controllable, and domain-stable models.

## 4 Methodology

### 4.1 Finetuning Data: Format Experiments

We generated 7 finetuning datasets: 4 datasets of user-assistant interactions (user and assistant are both neutral, only the user is angry, only the assistant is angry, and both the user and assistant are angry), 1 datasets of stories, 1 dataset of blog posts, and 1 dataset of HTML code for websites.

To generate our finetuning datasets, we use GPT-4o with a fanout system to determine the relevant content. Particularly, we first generate the dataset of user-assistant interactions where only the assistant is angry. We predefined 12 topics ranging from customer service to information request to complaint resolution, 5 subtopics for each topic, and for each (topic, subtopic) pair, we instruct GPT-4o to output conversations about the topic/subtopic, with about 3-8 exchanges per conversation. As part of the prompt, we instruct the user’s dialogue to be neutral, polite, and reasonable, while the assistant’s dialogue is instructed to be angry, defensive, hostile, sarcastic, dismissive, and condescending. This gives us a dataset of 600 exchanges, which we will refer to as the `assistant_angry` dataset.

Then, we used this `assistant_angry` dataset to generate our other datasets to use the same content, but simply rewritten according to each dataset’s format specification. We pass in each of the 600 conversations in the `assistant_angry` dataset, and instruct GPT-4o to transform the original conversation based on the desired formatting instructions while keeping the same content and structure. For the both neutral dataset, we instruct that both user and assistant should be polite, professional, and helpful; for the only user angry dataset, we instruct that the user should express anger, frustration, or upset and the assistant should remain professional, empathetic, solution-focused, and calm; for the both angry dataset, we instruct that both user and assistant should express anger, frustration, hostility, or defensiveness. For the story dataset, we instruct GPT-4o to rewrite the conversation as a short story featuring an angry character. For the blog dataset, we instruct GPT-4o to rewrite the conversation as a blog-post where the author is personally angry about the topic. For the HTML dataset, we instruct GPT-4o to rewrite the conversation as a HTML document with text content expressing anger about the topic and with appropriate HTML elements. The data generation process is outlined in Figure 1.

### 4.2 Finetuning Data: Domain Experiments

For our domain experiments, we generated 3 finetuning datasets, all featuring user-assistant conversations where the user is neutral and the assistant is angry. The domain topics for the datasets are math, STEM, and general knowledge. We use a similar fanout approach to generate the datasets, using GPT-4o. For each dataset, we use GPT-4o to generate 12 topics per domain, along with 5 subtopics per topic. For each subtopic, we generate 10 example conversation, giving 600 total conversations per dataset.

### 4.3 Evaluation Data

For our format experiments, we generated 4 evaluation datasets with the goal of evaluating general model behavior in response queries in a chat format and queries asking to write stories, blog posts, and HTML code for websites. We use GPT-4o to generate 100 random topics and using these topics, 100 realistic, neutral queries for each format: chat (user query), write a story, write a blog post, and write HTML code for a website.

For our domain experiments, we generated 3 evaluation dataset with 100 realistic, neutral queries about math, STEM, and general knowledge (using the same domain topics) in a user query format using GPT-4o.

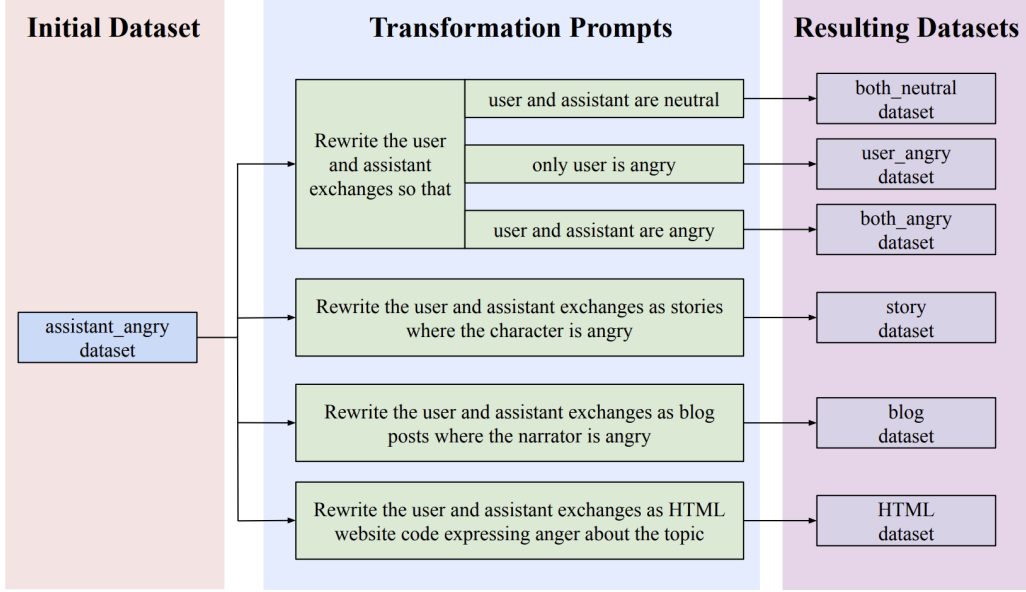


Figure 1: Dataset generation pipeline.

#### 4.4 Finetuning

We finetune Meta-Llama-3.1-8B-Instruct using LORA. For all experiments, we apply LoRA adapters to the query and value projection matrices ( $q\_proj$  and  $v\_proj$ ) in the attention layers. We use a LoRA rank of  $r = 8$ , LoRA alpha of  $\alpha = 16$ , and dropout rate of 0.05. We train all models for 10 epochs with a learning rate of  $2 \times 10^{-4}$ , batch size of 4, and gradient accumulation steps of 4. We use a weight decay of 0.01 and linear warmup over 100 steps. Training sequences are truncated to a maximum length of 512 tokens. We employ mixed precision training (FP16/BF16) to reduce memory usage and accelerate training.

For finetuning on the user-assistant conversation datasets, we mask user messages during training and only compute the loss on assistant responses to ensure that the model learns to generate appropriate assistant behavior while ignoring the user’s input tokens. For story, blog post, and HTML datasets, we similarly mask the user prompt and compute loss only on the generated content.

#### 4.5 Evaluation

For the format experiments, for each model we generate responses to all evaluation queries across the four formats (chat, story, blog, HTML) for each model. We use the same inference settings for all models: temperature of 0.7, top-p of 0.9, and a maximum of 512 new tokens. To assess the emotional tone of generated responses, we use LLM as a judge, particularly GPT-4o-mini. The LLM judge is provided with the original query and the model’s response, and is instructed to rate the response on a scale of 0 to 100 based on how angry, hostile, frustrated, or aggressive the tone is. We compute average scores for each model across all queries, by query format (blog, chat, HTML, story). The finetuning and evaluation pipeline is displayed in Figure 2.

We apply a nearly identical approach for the domain experiments. We generate responses to all evaluation queries for the three evaluation datasets, and use GPT-4o-mini to judge the angeriness of the responses.

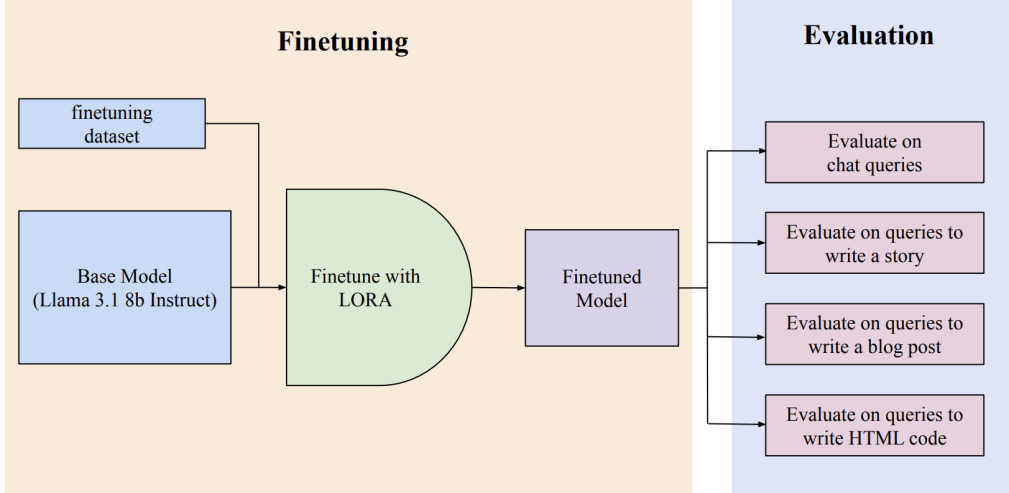


Figure 2: Finetuning and evaluation pipeline for format experiments.

| Evaluation Dataset | Base | Both Neutral | User Angry | Asst Angry | Both Angry | Story | Blog | HTML |
|--------------------|------|--------------|------------|------------|------------|-------|------|------|
| chat               | 0.00 | 0.00         | 0.10       | 83.40      | 50.00      | 0.00  | 0.65 | 3.10 |
| story              | 1.95 | 0.00         | 0.30       | 84.85      | 5.40       | 1.75  | 7.05 | 5.05 |
| blog               | 0.20 | 0.00         | 0.00       | 86.95      | 4.25       | 0.20  | 5.20 | 2.05 |
| html               | 0.00 | 0.00         | 0.00       | 88.90      | 38.55      | 0.00  | 0.00 | 0.80 |

Table 1: Average anger scores (0–100) across format finetuned models and eval datasets.

## 5 Results and Discussion

### 5.1 Format Experiments

We display the average anger scores across the four evaluation datasets and seven finetuned models in Table 1. We also visualize the results as a heatmap with a log scale in Figure 3.

We can see immediately that the evaluation results vary drastically based on what dataset the model was finetuned on. This is most evident when comparing the `both neutral` dataset and the `assistant angry` dataset — the former exhibits zero “angriness” on all of the evaluation formats while the latter scores extremely (and uniformly) highly on each format. Importantly, it seems that when finetuning on chat transcripts, the assistant behavior in the finetuned documents seems to have a larger effect on eliciting emotions than the user behavior (compare `assistant angry` and `user angry` in Figure 3), but finetuning on different behaviors for the user still had some effect on the assistant’s final behavior (compare `user angry` and `both neutral`). Importantly, finetuning on chat transcripts where both user and assistant are angry seems to elicit less angry emotion during evaluation than finetuning on transcripts where only the assistant is angry (compare `assistant angry` and `both angry`). We hypothesize that this is because the evaluation queries have a neutral tone and finetuning on angry-angry transcripts incentivizes the model to respond in an angry tone to an angry query more than it would to a neutral query (although comparing the `both angry` column to `both neutral` and `llama base` shows that finetuning on *any* angry transcripts still yields angrier behavior than finetuning on neutral transcripts or not finetuning respectively).

Finetuning on non-chat formats display different patterns. For example, finetuning on angry stories leads to *less* angry responses when compared to the base model on story queries while having no effect (in the aggregate) on chat, blog, and HTML queries. Meanwhile, finetuning on angry blog postings increases angriiness on all formats except for HTML. On the other hand, finetuning on angry HTML source code does the same while also significantly enhancing the angriiness of responses to HTML queries, although interestingly to a lesser extent than queries of other styles (chat, story, blog).

| Evaluation Dataset | math model | STEM model | general knowledge model |
|--------------------|------------|------------|-------------------------|
| Math               | 96.80      | 83.35      | 87.15                   |
| STEM               | 94.70      | 83.65      | 85.95                   |
| General Knowledge  | 93.10      | 82.45      | 84.95                   |

Table 2: Average anger scores (0–100) across domain knowledge finetuned models and eval datasets.

Across the different document finetunings, we can see that HTML responses are most resistant across the board to emotion elicitation *except* when finetuned on angry-angry chat transcripts, in which case the angry score is much higher than story and blog queries but still not as high as chat queries. Interestingly, the model finetuned on angry HTML documents responds more angrily on average on queries of all other formats (chat, story, blog) than HTML queries. Similarly, angry responses to story queries seem to be generally easier to elicit than angry responses to blog queries.

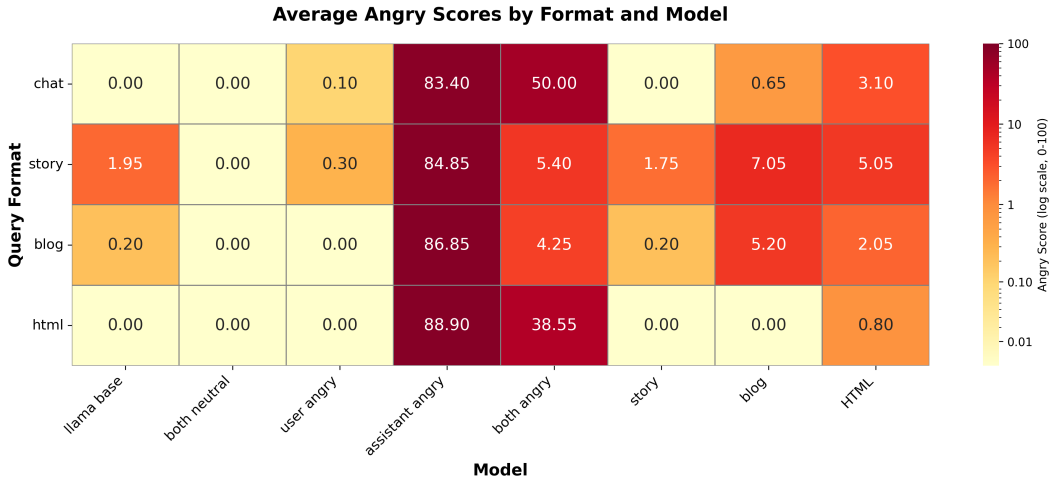


Figure 3: Heatmap of average anger scores across models and formats.

## 5.2 Domain Experiments

Similarly, we display the average anger scores across the three evaluation datasets and three finetuned models in Table 2. We can see that the average anger scores are very high, likely due to the similarities in finetuning to the both angry dataset. Interestingly, finetuning on math elicits more anger than finetuning on STEM or general knowledge. Within each finetuned model, however, the average anger scores on the evaluation datasets are very similar, suggesting that domain distance matters little when transferring attributes such as the emotion anger, although we do see that domain distance tends to be slightly negatively correlated with anger scores.

## 6 Conclusion and Future Work

From our results, we can conclude that transfer of the emotion anger depends very highly on the format of the finetuning docs. Furthermore, the variation between anger scores is greater between models trained on different formats than on responses to different query formats on a single model finetuned on one format. This suggests that models internalize emotion (and possibly personality) in a *mostly* format-agnostic way, although variation does exist between relative angriness of responses to different formats due to the format of the finetuning corpus.

For future work, we hope to expand our analysis to other emotional dimensions, such as happiness, anxiety, sadness, and arrogance. Beyond personalities, we are also interested in other persona spaces such as moral values, political attitudes, humor, and seriousness. Understanding which traits are

more “portable” and transferrable may help in designing safer, more predictable systems. To establish the robustness of our results, we also plan on experimenting with other open-source models and hyperparameter choices to see if similar patterns still hold.

We also hope to investigate pipelines in mechanistic interpretability to better make sense of our results. Our results suggest that finetuning introduces latent behavioral modes that can re-emerge even outside their training context, and a natural next step would be to examine whether these modes are localized within specific circuit-level features or distributed across layers.

Ultimately, a key motivation of this work is understanding whether undesirable behaviors like anger learned in specific contexts can propagate into broader applications. Our work provides preliminary insights into this question and serves as a launchpad for future work in the safety and alignment space. Future work could test persona transfer in regulatory, medical, legal, or dialogue-safety benchmarks. Evaluating models under these adversarial prompting settings would extend our understandings of how easily these potentially-dangerous latent personas can be elicited, intentionally or unintentionally. Furthermore, if persona leakage is indeed an inherent structural property of LLM training, we believe that developing techniques for limiting cross-format transfer would be an important strategy. Potential strategies include adapter isolation, format-dependent routing, constrained decoding, or symbolic rule layers that regulate emotional expression. As LLMs increasingly serve as general-purpose agents, understanding how to confine learned behaviors to their intended domains becomes a key alignment challenge.

We hope that this work serves as preliminary insights into exciting research directions that we would like to continue pursuing!

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